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Hartford, CT

PROFESSIONAL SUMMARY

Cybersecurity engineer and MS Cybersecurity candidate at Florida International University (GPA 3.89/4.0, expected May 2026) with end-to-end experience across SOC operations, network security engineering, vulnerability assessment, and offensive testing. CompTIA Security+ (SY0-701, DoD 8570/8140 IAT II baseline), eJPT, CRTA, CCSP-AWS (Merit), and CNSP (Merit) certified. Deployed full ELK Stack SIEM pipeline (Logstash → Elasticsearch → Kibana) achieving 95% VPN tunnel detection accuracy with custom DPI signatures and cutting MTTR from 30 min to under 8 min (70%+ improvement). Configured Cisco Secure Firewall 3100 across 3 enterprise sites reducing unauthorized access 40% and false-positive alerts 20%. Executed full OWASP Top 10 attack lifecycle against DVWA / Juice Shop at 95%+ exploitation success rate using Burp Suite, SQLMap, Hydra, and Metasploit; built PentestFlow and Automated Network Enumerator (Python) cutting recon time 30 to 40 percent across 50+ hosts. Author of a peer-reviewed paper on DPI-based VPN detection (2024). Teaching Assistant for 60+ FIU students across cybersecurity, networking, and SQL Server labs with 95% positive feedback. 1st place national winner, Eraksha Cybersecurity Competition (India); selected for Amazon Application Security CTF (NAMER, 2025). Available full-time May 2026; F-1 OPT (no immediate sponsorship); open to relocation, remote, or hybrid roles anywhere in the U.S.

CORE COMPETENCIES

SOC Operations & Threat Detection

SIEM Engineering (ELK Stack)

Incident Response (Full Lifecycle)

Network Security Engineering

Firewall Engineering (Cisco Secure)

IDS / IPS (Snort · Suricata)

Deep Packet Inspection · VPN Detection

Vulnerability Assessment (Nessus / OpenVAS)

Penetration Testing (eJPT · CRTA)

OWASP Top 10 Exploitation

Burp Suite · SQLMap · Metasploit

Active Directory & RBAC

AWS Cloud Security (IAM / GuardDuty)

Python / Bash / PowerShell Automation

MITRE ATT&CK · NIST CSF · ISO 27001

Security Documentation & Runbooks

WORK EXPERIENCE

Florida International University Aug 2025 – May 2026

Graduate Teaching Assistant — Database & SQL Server Administration · Miami, FL

- Led hands-on labs for **60+ students** on SQL Server installation, configuration, Always On AG / mirroring, backup & recovery, query plan analysis, and role-based access control; raised lab performance metrics **20%**.
- Authored concise admin runbooks — adopted as standard course reference; **95% positive student feedback**; reduced lab setup errors and student blockers across the semester.

Florida International University Jan 2025 – Aug 2025

Graduate Teaching Assistant — Cybersecurity & Networking Labs · Miami, FL

- Delivered weekly labs on Wireshark, Snort, Suricata, SIEM log analysis, and MITRE ATT&CK for **60+ students**; lifted lab completion **22%** across two semesters with **95% positive feedback**.
- Built scenario-based incident response assessments — packet injection, misconfigured SSH, unauthorized port scans — aligned to the IR lifecycle; cut grading turnaround **40%** via custom Python / Bash automation.
- Authored MITRE ATT&CK technique guides, IDS rule-writing walkthroughs, and Linux / iptables troubleshooting playbooks used as standard reference materials.

Network Security Analyst · Chennai, India

- Configured **Cisco Secure Firewall 3100** across 3 enterprise sites — zone-based policies, NAT, and ACLs cut unauthorized access incidents **40%**; consolidated 25+ redundant rules and reduced false-positive alerts **20%**.
- Standardized TLS 1.2 / 1.3 across endpoints and decommissioned legacy SSL / TLS 1.0, shrinking MITM and downgrade attack surface; ran VPN monitoring, user access reviews, and hardening across all managed segments.
- Automated 12+ Linux / Windows admin tasks via Bash, Python, and PowerShell — saved **6–8 hrs/week** and lifted operational efficiency **25%**; alert-driven response scripts (IP block, session termination) improved **MTTR 35%**.
- Performed traffic analysis (Wireshark, tcpdump, Nmap), contributed to centralized log monitoring and weekly incident summaries, and authored firewall change-management runbooks and TLS configuration standards.

PUBLICATION & SELECTED PROJECTS

VPN Blocker — Detection & Prevention of VPN Tunneling in Enterprise Networks Peer-Reviewed Publication · 2024

Published research paper on detecting unauthorized VPN tunnels (OpenVPN, WireGuard, HTTPS-obfuscated) without decryption — **90–95% detection accuracy** via flow metadata, handshake timing, TLS certificate characteristics, and packet-size entropy. Enforcement layer auto-triggered firewall ACL updates and session termination, cutting VPN policy violations **50–60%**; ELK SIEM integration with RBAC-routed alerts (L1/L2/L3) reduced **MTTR 30–40%**; complete audit trail supported compliance reviews.

Deep Packet Inspection · ELK Stack · RBAC · Network Policy Enforcement · Compliance & Audit

Home Networking Lab — DPI VPN Detection & ELK SIEM May–Jul 2025

Built a DPI engine classifying VPN tunnels (OpenVPN, WireGuard, HTTPS-obfuscated) at **95% accuracy** across 100+ sessions using packet-size fingerprints, inter-arrival timing, and TLS handshake anomalies. Full ELK pipeline (Logstash → Elasticsearch → Kibana) with automated alerting reduced MTTR from 30 min to **<8 min (70%+ improvement)**; RBAC restricted dashboard access by role. Cut VPN policy violations **60%** over a 10-week test.

Elasticsearch · Logstash · Kibana · Python · RBAC · Deep Packet Inspection

Cyber Range Lab — Cloud-Deployed Pentest Environment Cloud VPS · Apr 2026

Stood up a production-grade pentest lab on a cloud VPS — Bash IaC cut provisioning from 30+ min to **<5 min**. Ran the full OWASP Top 10 attack lifecycle (SQLi, XSS, command injection, auth bypass, credential brute-force) against DVWA and OWASP Juice Shop at **95%+ exploitation success rate**. UFW zone-based firewall enforced segmentation between Docker container networks and the public internet.

Docker · UFW · Kali Linux · Burp Suite · SQLMap · Hydra · Nmap · Nikto · Bash IaC

PentestFlow — Modular Automated Pentest Pipeline Jan–Feb 2026

Plugin-style automated assessment tool covering recon, host discovery, port scanning, service enumeration, CVE correlation, and structured HTML / Markdown report generation. Validated across **50+ host environments**; reduced manual effort **40%** and total assessment time **30%**.

Python · Nmap · CVE Correlation · Bash · Multi-threaded Scanning

Automated Network Enumerator Mar–Apr 2026

Python multi-threaded discovery tool — ICMP / ARP host sweep, TCP SYN scan (1–65535), banner grabbing, JSON output. Cut enumeration time **40%**; designed for direct hand-off into PentestFlow and the Metasploit workspace.

Python · Multi-threading · Nmap · Service Fingerprinting · JSON

EDUCATION

Master of Science — Cybersecurity

Aug 2024 – May 2026

Florida International University, Miami, FL — GPA 3.89 / 4.0 — Coursework: Network Security, Applied Cryptography, Digital Forensics & IR, Secure Software Development, Cyber Threat Intelligence, Cloud Security Architecture — Research: DPI-based VPN detection, automated pentest workflows, SIEM threat detection engineering

Bachelor of Engineering — Computer Science & Engineering

Nov 2020 – Mar 2024

Sri Krishna College of Engineering & Technology (Anna University), Tamil Nadu, India — Graduated with Distinction — Final project: real-time network intrusion detection system in Python / Scapy with rule-based detection and lightweight alert dashboard

CERTIFICATIONS & AWARDS

CompTIA Security+ (SY0-701) — CompTIA — DoD 8570/8140 IAT II baseline	May 2026
eJPT (Junior Penetration Tester) — INE Security	May 2026
Certified Red Team Analyst (CRTA) — CyberWarfare Labs	May 2026
INE Certified Cloud Associate (ICCA) — INE Security	May 2026
Certified Cloud Security Practitioner — AWS (CCSP-AWS) — Merit — The SecOps Group	Jan 2026
Certified Network Security Practitioner (CNSP) — Merit — The SecOps Group	Oct 2025
Amazon AppSec CTF — NAMER Region (Selected, 2025) Eraksha Cybersecurity Competition — 1st Place National, India (2023)	2025 / 2023

TECHNICAL SKILLS

SOC & Detection: ELK Stack (Elasticsearch, Logstash, Kibana — full pipeline), Splunk (familiar), Wazuh, Snort, Suricata, MITRE ATT&CK, alert engineering, L1/L2/L3 triage, threat hunting, incident response lifecycle, post-incident analysis

Network Security: Cisco Secure Firewall 3100, zone-based policies, ACLs, NAT, stateful inspection, OpenVPN, WireGuard, IPSec (IKEv1/IKEv2), DPI, Wireshark, tcpdump, TCP/IP (L2–L7), VLAN segmentation, subnetting

Offensive Security: Burp Suite, Metasploit Framework, SQLMap, Hydra, Nikto, Nmap (NSE), Nessus, OpenVAS, Netcat, OWASP Top 10, recon / enumeration / exploitation / post-exploitation, privilege escalation (Linux & Windows), credential harvesting

Cloud & IAM: AWS — IAM (least-privilege policy design, MFA), CloudTrail, GuardDuty, Security Hub, S3 security, VPC (security groups, NACLs, flow logs); Active Directory, Group Policy, RBAC; shared responsibility model; CIS AWS Benchmarks

Scripting & Automation: Python (tool dev, parsing, API calls), Bash / Shell (cron, log rotation, IaC), PowerShell (Windows automation, AD queries), SQL (log analysis), JavaScript (web-layer testing)

Frameworks & Compliance: NIST CSF, NIST 800–53, MITRE ATT&CK (Enterprise & Mobile), OWASP Top 10, CIS Benchmarks (Linux, Windows, AWS), ISO 27001 (awareness), SOC 2, PCI-DSS, HIPAA awareness

Operating Systems: Linux — Debian, Kali, Ubuntu Server, CentOS (hardening, systemd, fail2ban, rsyslog); Windows Server 2019 (Active Directory, Group Policy, Event Viewer); Windows 10 / 11 endpoint security

DevOps & Tooling: Git / GitHub (CI/CD with GitHub Actions), Docker (containerized labs, Dockerfile authoring), systemd (service units, watchdog timers), Bash automation, observability via ELK